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Continuous Time Linear Invariant System

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

53	27	1988 – 2017	English
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Answer key	at the end

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Marks 1 20 questions

Q1. GATE ECE 2017 Set 2

The input x(t) and the output y(t) of a continuous time system are related as $y(t) = \int_{t-T}^{t} x(u)du$.. The system is

(A) Linear and time variant
(B) linear and time invariant
(C) non linear and time variant
(D) nonlinear and time invariant

Open solution

Q2. GATE ECE 2017 Set 1

Consider the following statements for continuous-time linear time invariant (LTI) system.
I. There is no bounded input bounded output (BIBO) stable system with a pole in the right half of the complex plane.
II. There is no causal and BIBO stable system with a pole in the right half of the complex plane.

Which one among the following is correct?

(A) Both I and II are true
(B) Both I and II are not true
(C) Only I is true
(D) Only II is true

Open solution

Q3. GATE ECE 2015 Set 3

The impulse response of an LTI system can be obtained by

(A) differentiating the unit ramp response
(B) differentiating the unit step response
(C) integrating the unit ramp response
(D) integrating the unit step response

Open solution

Q4. GATE ECE 2015 Set 1

The result of the convolution $x(-t) * \delta(-t - t_0)$ is

(A) $x(t + t_0)$ (B) $x(t - t_0)$ (C) $x(-t + t_0)$ (D) $x(-t - t_0)$

Open solution

Q5. GATE ECE 2013

Two system with impulse responses $h_1(t)$ and $h_2(t)$ are connected in cascade. Then the overall impulse response of the cascaded system is given by

(A) Product of $h_1(t)$ and $h_2(t)$.
(B) Sum of $h_1(t)$ and $h_2(t)$.
(C) Convolution of $h_1(t)$ and $h_2(t)$.
(D) Subtraction of $h_1(t)$ and $h_2(t)$

Open solution

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Q6. GATE ECE 2013

The impulse response of a system is h(t) = t u(t). For an input u(t - 1), the output is

(A) $\frac{t^2}{2}u(t)$

(B) $\frac{t(t-1)}{2}u(t-1)$

(C) $\frac{(t-1)^2}{2}u(t-1)$

(D) $\frac{t^2-1}{2}u(t-1)$

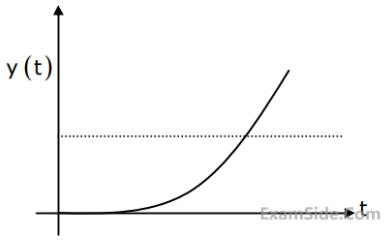
Open solution

Q7. GATE ECE 2011

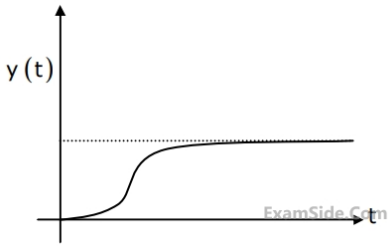
The differential equation $100\frac{d^2y}{dt} - 20\frac{dy}{dt} + y = x(t)$ describes a system with an input x(t) and output y(t). The system, which is initially relaxed, is excited by a unit step input. The output y(t) can be represented by the waveform

Open solution

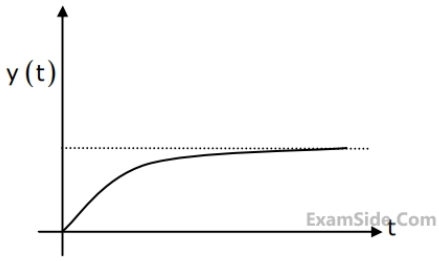
(A)



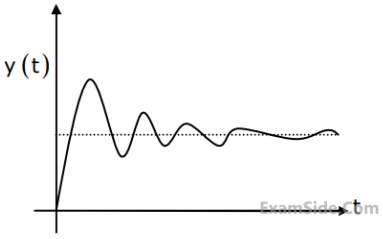
(B)



(C)



(D)



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Q8. GATE ECE 2008

The input and output of a continuous system are respectively denoted by x(t) and y(t). Which of the following descriptions corresponds to a causal system?

Open solution

(A) $y(t) = x(t-2) + x(t+4)$

(B) $y(t) = (t-4)x(t+1)$

(C) $y(t) = (t+4)x(t-1)$

(D) $y(t) = (t+5)x(t+5)$

Q9. GATE ECE 2008

The impulse response h(t) of a linear time-invariant continuous time system is described by $h(t) = \exp(\alpha t)u(t) + \exp(\beta t)u(-t)$, where u(t) denotes the unit step function, and α and β are real constants. This system is stable if

Open solution

(A) α is positive and β is positive

(B) α is negative and β is negative

(C) α is positive and β is negative

(D) α is negative and β is positive

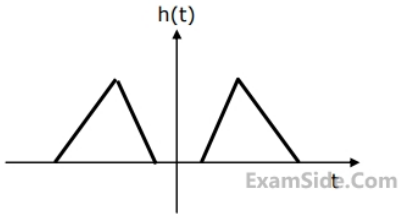
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Q10. GATE ECE 2005

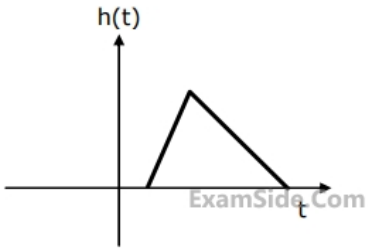
Open solution

Which of the following can be impulse response of a causal system?

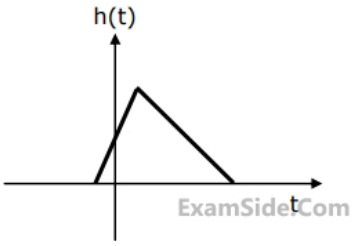
(A)



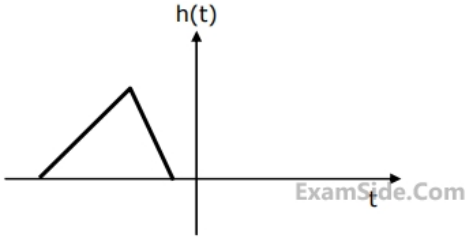
(B)



(C)



(D)



Q11. GATE ECE 2003

Open solution

Let x(t) be the input to a linear, time-invariant system. The required output is 4x(t - 2). The transfer function of the system should be

(A) $4\,e^{j4\pi f}$ (B) $2\,e^{-j8\pi f}$ v (C) $4\,e^{-j4\pi f}$ (D) $2\,e^{j8\pi f}$

Q12. GATE ECE 2002

Open solution

Convolution of x(t + 5) with impulse function $\delta\,(t\,-\,7)$ is equal to

(A) $X\,(t-12)$ (B) $X\,(t+12)$ (C) $X\,(t-2)$ (D) $X\,(t+2)$

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Q13. GATE ECE 2001

The transfer function of a system is given by $H(s) = \frac{1}{s^2(s-2)}$. The impulse response of the system is

(A) $\left(t^2 * e^{-2t}\right)U(t)$

(B) $\left(t * e^{2t}\right)U(t)$

(C) $\left(t^2 e^{-2t}\right)U(t)$

(D) $\left(te^{-2t}\right)U(t)$

Open solution

Q14. GATE ECE 2000

A system with an input x(t) and an output y(t) is described by the relation: y(t) = t x(t). This system is

(A) linear and time-invariant.

(B) linear and time-varying.

(C) non-linear and time-invariant.

(D) non-linear and time-varying.

Open solution

Q15. GATE ECE 1998

The unit impulse response of a linear time invariant system is the unit step function u(t). For t>0, the response of the system to an excitation e^{-at} u(t), a>0 will be

(A) $a\ e^{-at}$

(B) $(1/a)\ (1 - e^{-at})$

(C) $a(1 - e^{-at})$

(D) $1 - e^{-at})$

Open solution

Q16. GATE ECE 1998

The transfer function of a zero - order - hold system is

(A) $(1/s)\left(1 + e^{-sT}\right)$v

(B) $(1/s)\left(1 - e^{-sT}\right)$

(C) $1 - (1/s)e^{-sT}$

(D) $1 + (1/s)e^{-sT}$

Open solution

Q17. GATE ECE 1995

The transfer function of a linear system is the

(A) ratio of the output, $v_0(t)$, and input, $v_i(t)$.

(B) ratio of the derivatives of the output and the input.

(C) ratio of the Laplace transform of the output and that of the input with all initial conditions zeros.

(D) none of these.

Open solution

Q18. GATE ECE 1995

Non - minimum phase transfer function is defined as the transfer function

(A) which has zeros in the right - half s - plane.

(B) which has zeros only in the left - half s - plane.

(C) which has poles in the right - half s - plane.

(D) which has poles in the left - half s - plane.

Open solution

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Q19. GATE ECE 1995

Open solution

Let h(t) be the impulse response of a linear time invariant system. Then the response of the system for any input u(t) is

(A) $\int_0^t h(\tau)u(t-\tau)d\tau$

(B) $\frac{d}{dt}\int_0^t h(\tau)u(t-\tau)d\tau$

(C) $\int_0^t \left[\int_0^t h(\tau)u(t-\tau)d\tau\right]dt$

(D) $\int_0^t h^2(\tau)u(t-\tau)d\tau$

Q20. GATE ECE 1994

Open solution

Indicate whether the following statement is TRUE/FALSE: Give reason for your answer.
If G(s) is a stable transfer function, then $F(s) = \frac{1}{G(s)}$ is always a stable transfer function.

(A) TRUE (B) FALSE

Marks 2 29 questions

Q21. GATE ECE 2017 Set 2

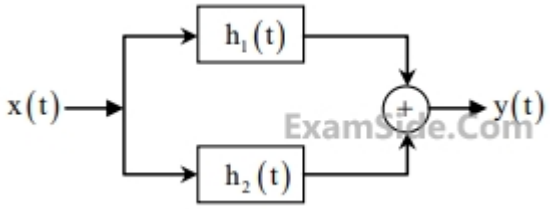
Open solution

The transfer function of a causal LTI system is H(s) = 1/s. If the input to the system is x(t) = $[\sin(t)/\pi t]u(t)$; where u(t) is a unit step function. The system output y(t) as $t \to \infty$ is _____.

Q22. GATE ECE 2017 Set 2

Open solution

Consider the parallel combination of two LTI systems shown in the figure.



The impulse responses of the systems are

$h_1(t) = 2\delta(t+2) - 3\delta(t+1)$
$h_2(t) = \delta(t-2)$
If the input x(t) is a unit step signal, then the energy of y(t) is_____.

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Q23. GATE ECE 2017 Set 2

Open solution

Consider an LTI system with magnitude response

$$|H(f)| = \begin{cases} 1 - \frac{|f|}{20}, & |f| \leq 20 \\ 0, & |f| > 20 \end{cases}$$

and phase response Arg[H(f)]= - 2f.
If the input to the system is $x(t) = 8\cos\left(20\pi t + \frac{\pi}{4}\right) + 16\sin\left(40\pi t + \frac{\pi}{8}\right) + 24\cos\left(80\pi t + \frac{\pi}{16}\right)$
Then the average power of the output signal y(t) is_____.

Q24. GATE ECE 2017 Set 1

Open solution

A continuous time signal x(t) = $4\cos(200\pi t) + 8\cos(400\pi t)$, where t is in seconds, is the input to a linear time invariant (LTI) filter with the impulse response $h(t) = \begin{cases} \frac{2\sin(300\pi t)}{\pi t}, & t \neq 0 \\ 600, & t = 0 \end{cases}$
Let y(t) be the output of this filter. The maximum value of $|y(t)|$ is _____.

Q25. GATE ECE 2015 Set 2

Open solution

The output of a standard second-order system for a unit step input is given as $y(t) = 1 - \frac{2}{\sqrt{3}}e^{-t}\cos\left(\sqrt{3}t - \frac{\pi}{6}\right)$.
The transfer function of the system is

(A) $\frac{2}{(s+2)(s+\sqrt{3})}$
(B) $\frac{1}{s^2+2s+1}$
(C) $\frac{3}{s^2+2s+3}$
(D) $\frac{4}{s^2+2s+4}$

Q26. GATE ECE 2015 Set 2

Open solution

Input x(t) and output y(t) of an LTI system are related by the differential equation y"(t) - y'(t) - 6y(t) = x(t). If the system is neither causal nor stable, the imulse response h(t) of the system is

(A) $\frac{1}{5}e^{3t}u(-t) + \frac{1}{5}e^{-2t}u(-t)$
(B) $\frac{-1}{5}e^{3t}u(-t) + \frac{1}{5}e^{-2t}u(-t)$
(C) $\frac{1}{5}e^{3t}u(-t) + \frac{1}{5}e^{-2t}u(t)$
(D) $\frac{-1}{5}e^{3t}u(-t) - \frac{1}{5}e^{-2t}u(t)$

Q27. GATE ECE 2013

Open solution

The impulse response of a continuous time system is given by $h(t) = \delta(t-1) + \delta(t-3)$. The value of the step response at t = 2 is

(A) 0 (B) 1 (C) 2 (D) 3

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Q28. GATE ECE 2012

The input x(t) and output y(t) of a system are related as y(t) = $\int_{-\infty}^{t} x(\tau)\cos(3\tau)d\tau$.

The system is

(A) time-invariant and stable.
(B) stable and not time-invariant.
(C) time-invariant and not stable.
(D) not time-invariant and not stable.

Open solution

Q29. GATE ECE 2011

An input x(t) = exp(-2t) u(t) + δ(t-6) is applied to an LTI system with impulse response h(t) = u(t). The output is

(A) [1- exp(-2t)] u(t) + u(t+6)
(B) [1- exp(-2t)] u(t) + u(t-6)
(C) 0.5 [1 - exp(-2t)] u(t) + u(t+6)
(D) 0.5 [1- exp(-2t)] u(t) + u(t-6)

Open solution

Q30. GATE ECE 2010

A continuous time LTI system is described by $\frac{d^2y(t)}{dt^2} + 4\frac{dy(t)}{dt} + 3y(t) = 2\frac{dx(t)}{dt} + 4x(t)$.
Assuming zero initial conditions, the response y(t) of the above system for the input x(t) = e^{-2t} u(t) is given by

(A) $(e^{t} - e^{3t})\,u(t)$ (B) $(e^{-t} - e^{-3t})\,u(t)$ (C) $(e^{-t} + e^{-3t})\,u(t)$ (D) $(e^{t} + e^{3t})\,u(t)$

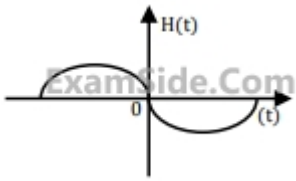
Open solution

Q31. GATE ECE 2009

Consider a system whose input x and output y are related by the equation

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau)\,h(2\tau)\,d\tau$$

Where h(t) is shown in the graph.



Which of the following four properties are possessed by the system?

BIBO: Bounded input gives a bounded output.
Causal: The system is casual.
LP: The system is low pass.
LTI: The system is linear and time- invariant.

(A) Causal, LP
(B) BIBO, LTI
(C) BIBO, Causal, LTI
(D) LP, LTI

Open solution

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Q32. GATE ECE 2008

A linear, time-invariant, causal continuous time system has a rational transfer function with simple poles at s = - 2 and s = - 4, and one simple zero at s = - 1. A unit step u(t) is applied at the input of the system. At steady state, the output has constant value of 1. The impulse response of this system is

Open solution

(A) [exp(-2t) + exp(-4t)]u(t)

(B) [-4exp(-2t) + 12 exp(-4t) - exp(-t)]u(t)

(C) [-4 exp(-2t) +12 exp(-4t)]u(t)

(D) [-0.5 exp(-2t) +1.5 exp(-4t)]u(t)

Q33. GATE ECE 2008

Let x(t) be the input and y(t) be the output of a continuous time system. Match the system properties P1, P2 and P3 with system relations R1, R2, R3, R4.

Open solution

Properties

P1 : Linear but NOT time-invariant
P2: Time-invariant but NOT linear
P3: Linear and time-invariant

Relations

R1: y(t) = t^2 **x(t)**
R2: y(t) = t$|x(t)|$
R3: y(t) = $|x(t)|$
R4: y(t) = x(t-5)

(A) (P1, R1), (P2, R3), (P3, R4)

(B) (P1, R2), (P2, R3), (P3, R4)

(C) (P1, R3), (P2, R1), (P3, R2)

(D) (P1, R1), (P2, R2), (P3, R3)

Q34. GATE ECE 2007

The frequency response of a linear, time-invariant system is given by H(f) = $\frac{5}{1+j10\pi f}$. The step response of the system is

Open solution

(A) $5(1-e^{-5t})\,u(t)$

(B) $5\left(1-e^{-\frac{t}{5}}\right)u(t)$

(C) $\frac{1}{5}\left(1-e^{-5t}\right)u(t)$

(D) $\frac{1}{5}\left(1-e^{-\frac{t}{5}}\right)u(t)$

Q35. GATE ECE 2006

Let g(t) = p(t) * p(t), where * denotes convolution and p(t) = u(t) - (t-1) with u(t) being the unit step function. The impulse response of filter matched to the singal s(t) = g(t) - $[\delta(t-2)*g(t)]$ is given as

Open solution

(A) s(1 - t)

(B) -s (1 - t)

(C) -s(t)

(D) s(t)

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Q36. GATE ECE 2005

The output y(t) of a linear time invariant system is related to its input x(t) by the following equation: y(t) = 0.5 x $(t - t_d + T) + x\,(t - t_d) + 0.5\,x(t - t_d - T)$. The filter transfer function $H(\omega)$ of such a system is given by

Open solution

(A) $(1 + \cos \omega T)e^{-j\omega t_d}$

(B) $(1 + 0.5 \cos \omega T)e^{-j\omega t_d}$

(C) $(1 + \cos \omega T)e^{j\omega t_d}$

(D) $(1 - 0.5 \cos \omega T)e^{-j\omega t_d}$

Q37. GATE ECE 2004

A system described by the differential equation: $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = x(t)$ is initially at rest. For input x(t) = 2u(t), the output y(t) is

Open solution

(A) $(1 - 2e^{-t} + e^{-2t})\,u(t)$

(B) $(1 + 2e^{-t} - 2\,e^{-2t})\,u(t)$

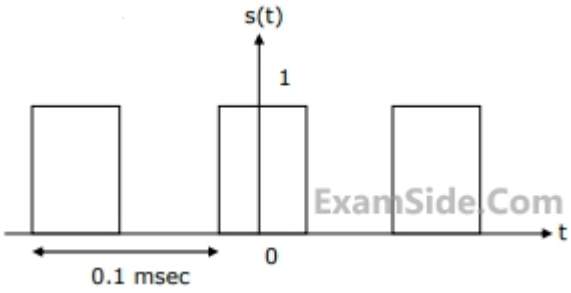
(C) $(0.5 + e^{-t} + 1.5\,e^{-2t})\,u(t)$

(D) $(0.5 + 2e^{-t} + 2\ e^{-2t})\,u(t)$

Q38. GATE ECE 2004

A rectangular pulse train s(t) as shown in Fig.1 is convolved with the signal $\cos^2\,(4\pi\,10^3\,t)$. The convolved signal will be a

Open solution



(A) DC

(B) 12 kHz sinusoid

(C) 8 kHz sinusoid

(D) 14 kHz sinusoid

Q39. GATE ECE 2004

A causal system having the transfer function H(s) = $\frac{1}{s+2}$, is excited with 10 u(t). The time at which the output reaches 99% of its steady state value is

Open solution

(A) 2.7 sec

(B) 2.5 sec

(C) 2.3 sec

(D) 2.1 sec

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Q40. GATE ECE 2001

Open solution

The impulse response function of four linear system S1, S2, S3, S4 are given respectively by h_1(t), = 1;

h_2(t), = U(t);

$h_3(t) = \frac{U(t)}{t+1}$;

$h_4(t) = e^{-3t}U(t)$,

where U (t) is the unit step function. Which of these system is time invariant, causal, and stable?

(A) SI (B) S2 (C) S3 (D) S4

Q41. GATE ECE 2000

Open solution

A linear time invariant system has an impulse response e^{2t}, $t > 0$. If the initial conditions are zero and the input is e^{3t}, the output for $t > 0$ is

(A) $e^{3t} - e^{2t}$ (B) e^{5t}

(C) $e^{3t} + e^{2t}$ (D) None of the above

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Q42. GATE ECE 2000

Open solution

Let u(t) be the unit step function. Which of the waveforms in Fig.(a) -(d) corresponds to the convolution of $[u(t) - u(t-1)]$ with $[u(t) - u(t-2)]$?

(A)

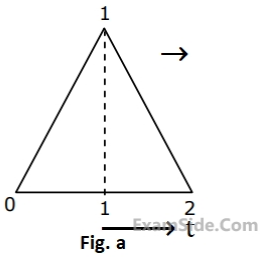


Fig. a

(B)

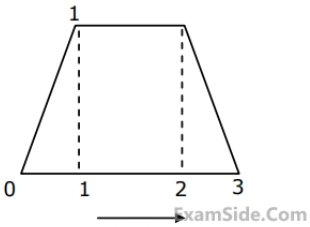


Fig. b

(C)

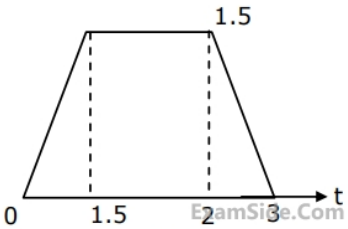


Fig. c

(D)

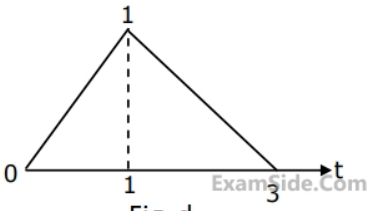


Fig. d

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Q43. GATE ECE 1997

Open solution

Match each of the items 1, 2 on the left with the most appropriate item A, B, C or D on the right. In the case of a linear time invariant system

List - 1

(1) Poles in the right half plane implies.

(2) Impulse response zero for $t \leq 0$ implies.

List - 2

(A) Exponential decay of output

(B) System is causal

(C) No stored energy in the system

(D) System is unstable

Q44. GATE ECE 1994

Open solution

Match each of the items A, B and C with an appropriate item from 1, 2, 3, 4 and 5.

List - 1

(A) $a_1 \frac{d^{2y}}{dx^2} + a_2 y \frac{dy}{dx} + a_3 y = a_4$

(B) $a_1 \frac{d^3 y}{dx^3} + a_2 y = a_3$

(C) $a_1 \frac{d_2 y}{dx_2} + a_2 x \frac{dy}{dx} + a_3 x^2 y = 0$

List - 2

(1) Non linear differential equation.

(2) Linear differential equation with constant coefficients.

(3) Linear homogeneous differential equation.

(4) Non - Linear homogeneous differential equation.

(5) Non - linear first order differential equation.

Q45. GATE ECE 1991

Open solution

An excitation is applied to a system at $t = T$ and its response is zero for $-\infty < t < T$. Such a system is a

(A) non-causal system

(B) stable system

(C) causal system

(D) unstable system

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Q46. GATE ECE 1991

The voltage across an impedance in a network is V(s) = Z(s) I(s), where V(s), Z(s) and I(s) are the Laplace Transforms of the corresponding time functions V(t), z(t) and i(t).
The voltage v(t) is

Open solution

(A) $v(t) = z(t) \,.\, i(t)$

(B) $v(t) = \int_0^t i(\tau)\, z(t-\tau) d\tau$

(C) $v(t) = \int_0^t i(\tau) z(t+\tau) d\tau$

(D) $v(t) = z(t) + i(t)$

Q47. GATE ECE 1990

The response of an initially relaxed linear constant parameter network to a unit impulse applied at $t = 0$ is $4e^{-2t}u(t)$. The response of this network to a unit step function will be:

Open solution

(A) $2\left[1 - e^{-2t}\right]u(t)$

(B) $4\left[e^{-t} - e^{-2t}\right]u(t)$

(C) $\sin 2t$

(D) $\left(1 - 4e^{-4t}\right)u(t)$

Q48. GATE ECE 1990

The impulse response and the excitation function of a linear time invariant casual system are shown in Fig. a and b respectively. The output of the system at t = 2 sec. is equal to

Open solution

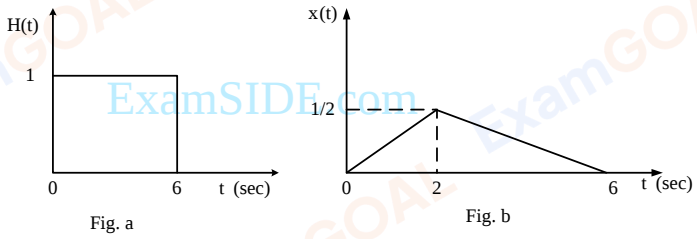


Fig. a

Fig. b

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(A) 0 (B) 1/2 (C) 3/2 (D) 1

Q49. GATE ECE 1988

The transfer function of a zero-order hold is

Open solution

(A) $\frac{1-\exp(-Ts)}{s}$

(B) $1/s$

(C) 1

(D) $1/\left[1 - \exp\left(-Ts\right)\right]$

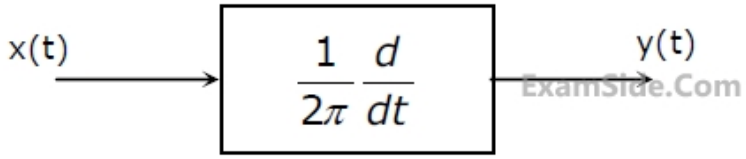
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Marks 5 4 questions

Q50. GATE ECE 2002

Open solution

A deterministic signal x(t) = $\cos(2\pi t)$ is passed through a differentiator as shown in Figure.



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(a) Determine the autocorrelation $R_{xx}(\tau)$ and the power spectral density $S_{xx}(f)$.

(b) Find the output power spectral density $S_{yy}(f)$.

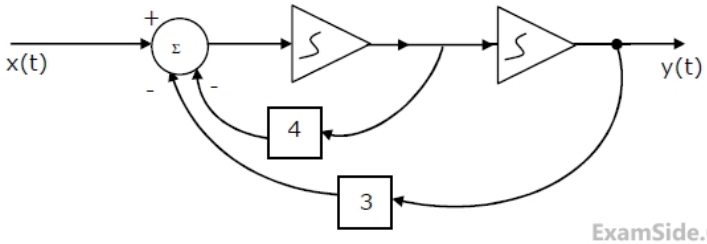
(c) Evaluate $R_{xy}(0)$ and $R_{xy}(1/4)$.

Q51. GATE ECE 2000

Open solution

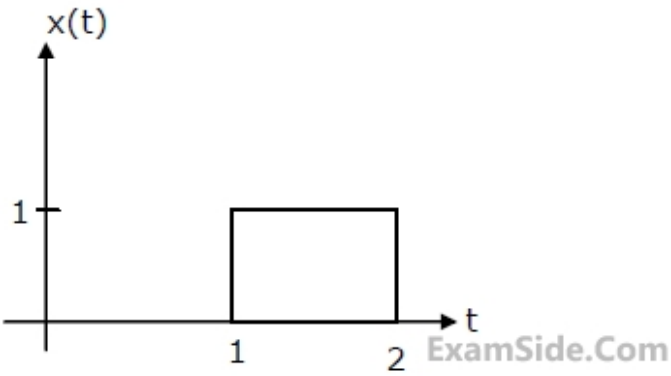
For the linear, time-invariant system whose block diagram is shown in Fig.(a), with input x(t) and output y(t).

(a) Find the transfer function.
(b) For the step response of the system [i.e. find y(t) when x(t) is a unit step function and the initial conditions are zero]
(c) Find y(t), if x(t) is as shown in Fig.(b), and the initial conditions are zero.



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Fig (a)



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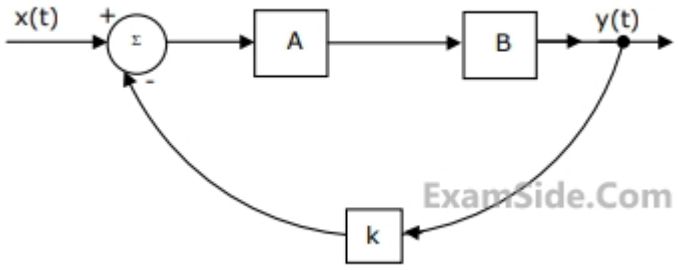
Fig (b)

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Q52. GATE ECE 1997

Open solution

Fig.1, shows the block diagram representation of a control system. The system in block A has an impulse response $h_A(t) = e^{-t}\,u(t)$. The system in block B has an impulse response $h_B(t) = e^{-2t}\,u(t)$. The block 'k' amplifies its input by a factor k. For the overall system with input x(t) and output y(t)



(a) Find the transfer function $\frac{Y(s)}{X(s)}$, when k=1

(b) Find the impulse response, when k = 0

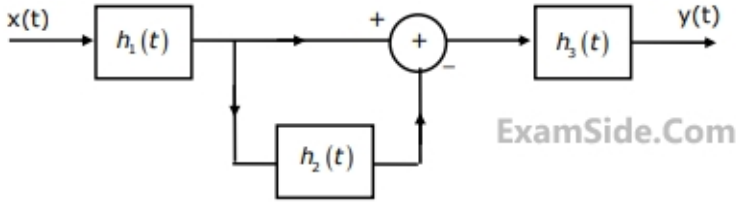
(c) Find the value of k for which the system becomes unstable.

$$\left[Note: u(t) \equiv 0 \quad t \leq 0 \right] \\ \equiv 1 \quad t > 0$$

Q53. GATE ECE 1993

Open solution

Consider the following interconnection of the three LTI systems (Fig.1). $h_1(t)$, $h_2(t)$ and $h_3(t)$ are the impulse responses of these three LTI systems with $H_1(\omega)$, $H_2(\omega)$, and $H_3(\omega)$ as their respective Fourier transforms. Given that $h_1(t) = \frac{d}{dt}\left[\frac{\sin(\omega_0 t)}{2\pi t}\right]$, $H_2(\omega) = \exp\left(\frac{-j2\pi\omega}{\omega_0}\right)$ $h_3(t) = u(t)\,and\,x(t) = \sin 2\omega_0 t + \cos(\omega_0 t/2)$, find the output y(t).



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Answer Key

Marks 1

1. **B**	2. **D**	3. **B**	4. **D**	5. **C**
6. **C**	7. **A**	8. **C**	9. **D**	10. **B**
11. **C**	12. **C**	13. **B**	14. **B**	15. **B**
16. **B**	17. **C**	18. **A**	19. **A**	20. **B**

Marks 2

21. **0.5**	22. **7**	23. **8**	24. **8**	25. **D**
26. **B**	27. **B**	28. **D**	29. **D**	30. **B**
31. **B**	32. **C**	33. **A**	34. **B**	35. **C**
36. **A**	37. **A**	38. **A**	39. **C**	40. **D**
41. **D**	42. **D**	43. **(1 - D), (2 - B)**	44. **(A - 1), (B - 2), (C - 4)**	45. **C**
46. **B**	47. **A**	48. **B**	49. **A**	

Marks 5

50. **(a) $R_{xx}(\tau)$ =** $\frac{1}{2}\cos(2\pi t)$ **$S_{xx}(f)$ =** $\frac{1}{4}[\delta(f-1)+\delta(f+1)]$ **(b) $S_{yy}(f)$ =** $\frac{1}{4}[\delta(f-1)+\delta(f+1)]$ **(c) $R_{xy}(0) = 0$ $R_{xy}(\frac{1}{4}) = \frac{1}{2}$**	51. **(a) Transfer Function =** $\frac{1}{(s+1)(s+3)}$ **(b) y(t) =** $\left[\frac{1}{3}-\frac{1}{2}e^{-1}+\frac{1}{6}e^{-3t}\right]u(t)$	52. **a)** $\frac{Y(s)}{X(s)}=\frac{1}{S^2+2S+3}$ **b) Impulse response =** $\left[e^{-t}-e^{-2t}\right]u(t)$ **c) System is unstable when k < -2**	53. **y(t) =** $\cos\left(\frac{w_0}{2}\right)t$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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